metaTextGrad: Automatically optimizing language model optimizers

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HarnessRSI 64 ★★

evidence: improver

Paper

Code

1 Optimize the optimizer

1 Seed optimizers
TGD: changes task-program prompts
ADAS-TG: changes program structure

▼

2 Meta prompt stage · o1
Training question + answer + optimizer
→ rewrite optimizer task description

▼

3 Inner optimization
Candidate optimizer edits task program
→ best program's validation accuracy

▼

4 Keep if better · per §3.3
Same validation set grades both levels
Init: evaluated input vs App. A zero score

▼

5 Meta structure stage · o1
Generate optimizer composition code
Repeat inner test + validation keep rule

Kept: optimizer prompts + composition code
Weights fixed; top-level meta-optimizer fixed

Task model: GPT-4o-mini. TGD/MIPRO: GPT-4o; structure/meta: o1. Structure stage combines existing strategies in Python, with fixed interfaces.

2 Average test: 0.53 vs 0.47

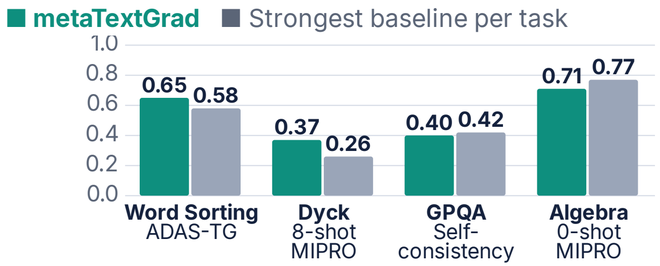


Table 1 · GPT-4o-mini · test means over five seeds (§4.1), no SD/CI. Average 0.47 is the zero-shot MIPROv2 row, not the per-task baseline envelope.

4 Where gains stop

- **BBH carries the gain.** GPQA 0.40 vs self-consistency 0.42; Algebra 0.71 vs MIPRO 0.77 (ties TGD).
- **Haiku Dyck:** test 0.35 vs ADAS 0.34; validation 0.32 vs 0.35. No SD/CI (Table 4).
- **Shared validation** at both levels; separate test splits, but no overlap/contamination audit.
- **AIME 2024:** no significant GPT-4o-mini gain; no scores supplied (Appendix I).

3 Controls & unresolved costs

Dyck ablation · test (Table 6)

Optimizer	Original	Meta-prompt
TGD	0.10	**0.24**
ADAS-TG	0.16	**0.37**
Structure only		0.16
Full metaTextGrad		0.37

Full = prompt-optimized ADAS-TG; no better composition found. No SD/CI. Prose/test scope unresolved: Algebra ties TGD; structure-only ties ADAS on test, not validation (0.21 → 0.24).

Task transfer

GPQA → Algebra: 0.77 vs TGD 0.71 (Table 5); in-domain MIPRO also 0.77. Target reset/data access unclear.

Budget ≠ six steps per epoch

TGD/MIPRO get GPT-4o; ADAS-TG gets o1, but no outer rewriting. Matching total budgets, baseline tuning budgets and hyperparameter-selection split unreported.

Nested cost · per meta epoch, Algebra

~400k program + ~100k optimizer + ~2.5k meta tokens (Table 2); not full search.
Dyck mini $0.44 / 0.37 vs zero-shot 4o $0.52 / 0.18 (Table 3); full-search amortization unclear.

Grading is not uniform

BBH: GPT-4o + reference. Algebra/GPQA: exact match, except DSPy uses GPT-4o. No judge audit; favoritism not proven.

Takeaway

Task-aligning optimizer prompts helps downstream search. The clearest ablation does not show extra benefit from composition. BBH gains do not extend to leading on the knowledge tasks, and equal full-search budgets remain unproven.

Our read · HarnessRSI

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Reports: test splits, transfer, ablations, seed averaging, per-epoch costs.
Not reported: SD/CI, matched total budgets, full-search bill, leakage audit.